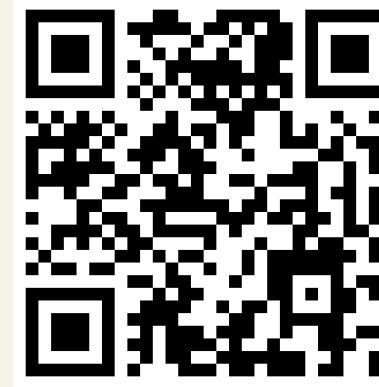


UEBAI: Universal Educational Bots — Virtual AI Robot Teachers That Read Topic Cards, Not the Whole Textbook

Student Name: Rashed Ahmed Saad Aseri

ID:



Introduction

Students increasingly ask AI assistants about their own school textbooks. The usual way is to upload the whole book, so the model re-reads hundreds of pages (about 100,000–190,000 tokens) with **every** question: answers are slow and costly, only large long-context models can do it, and models use facts from the middle of long texts less reliably (Liu et al.). Standard retrieval (RAG) instead cuts the book into fixed-length chunks that start and end mid-idea, and repeats itself when a student has several overlapping sources such as the textbook and the teacher's slides.

The aim of UEBAI is to design and test an Arabic-first web app in which a student builds a customisable 3D robot teacher from their own resources. The book is indexed **once** into topic cards, and each question is answered from one card with verified page citations. The objectives are to build a working prototype, compare it with reading the whole book on real Saudi Ministry of Education textbooks using answer keys written before any run, and test merging a second, overlapping source.

Research question: Can an AI teacher that reads one topic card answer as well as one that reads the whole book, while being faster and cheaper?

THE PROBLEM

1 Whole book, every question
≈160,000 tokens re-read each time

2 Slow and costly
Long waits; big models only

3 Overlapping sources
Slides and book repeat or disagree

UEBAI APPROACH

Index once
Book → topic cards

Read one card
≈3,000 tokens per question

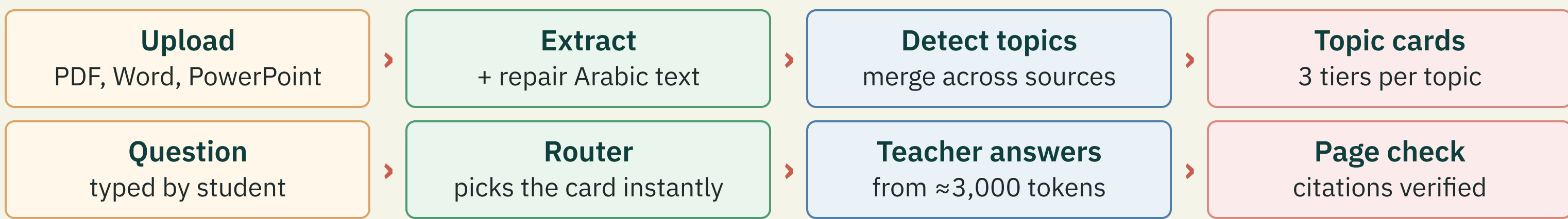
Verify pages
Invented pages are dropped

Topic cards • Merged sources • Verified pages • Arabic-first

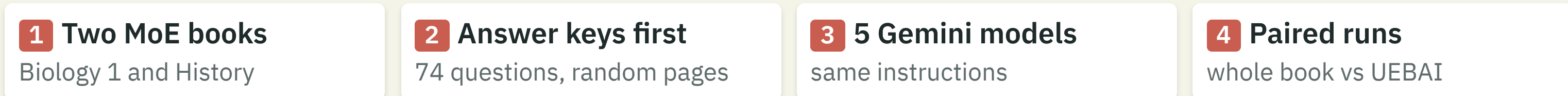
Fig. 1. Re-reading the whole book for every question is slow and costly; UEBAI indexes the book once and reads only the card a question needs.

Methodology

1 Architecture



2 Test plan



3 Measurements

Setup:	27 questions on up to 5 models gave 69 pairs: the same question on the same model, once with the whole book and once with UEBAI, in shuffled order. 7 questions were not in the book.
Measures:	Tokens read, time to first word, total time, key facts recalled (graded by code), correct page, and saying "not in the book".
Statistics:	Paired permutation test, 95% confidence interval, and a 10-point non-inferiority margin fixed in advance.
Fairness:	Answer keys were written before any run and fingerprinted (SHA-256); a free retrieval check covered all 66 answerable questions.

Results

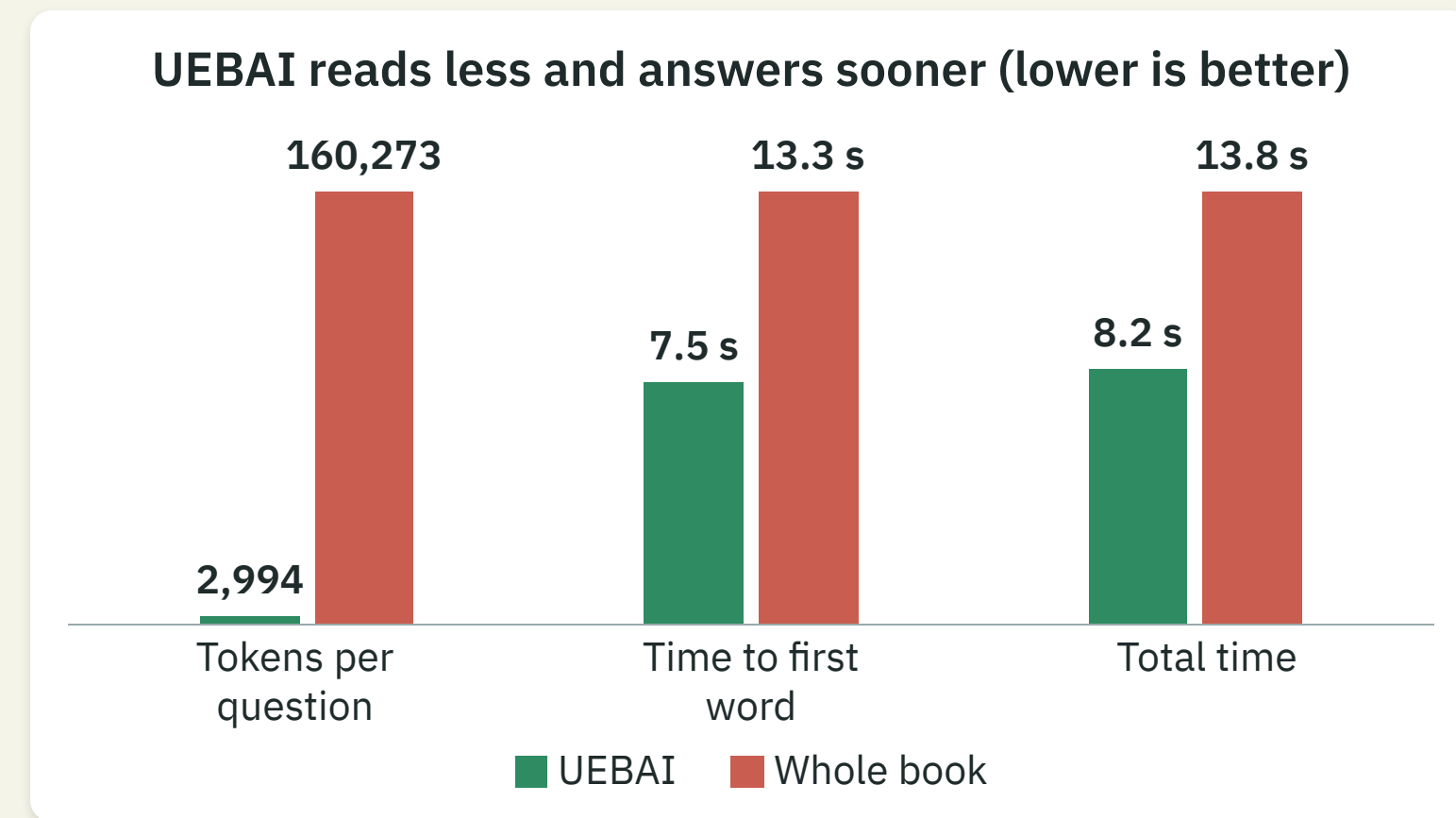


Fig. 2. UEBAI prototype: an answer with page citations and the cards the teacher read.

55× fewer tokens	2× faster first word	Not worse in answer quality	1.7% of the book read
Measure (69 pairs)	UEBAI	Whole book	Result
Tokens per question	2,994	160,273	55× fewer in 69/69, $p < 0.0001$
Time to first word	7.5 s	13.3 s	faster in 49/69, $p = 0.024$
Total time	8.2 s	13.8 s	faster in 48/69, $p = 0.037$
Key facts recalled	92%	97%	not worse (CI −9 to +1 pts)
Correct page cited	61/63	63/64	similar

Times are averages; the median speed-up was 2×. In the first biology batch the gain depended on the model: 28× on gemini-3-flash-preview, 3.8× on 3.5-flash, 2.4× on 3.7-flash and only 1.1× on 2.5-flash. In the free retrieval check, UEBAI's material held 95% of the biology answer facts while being only 1.7% of the book.

Innovation

UEBAI's unit of knowledge is the **study topic**, not a fixed-length chunk. It builds on retrieval-augmented generation (Lewis et al.); what is new is how the parts are designed and combined for textbooks.

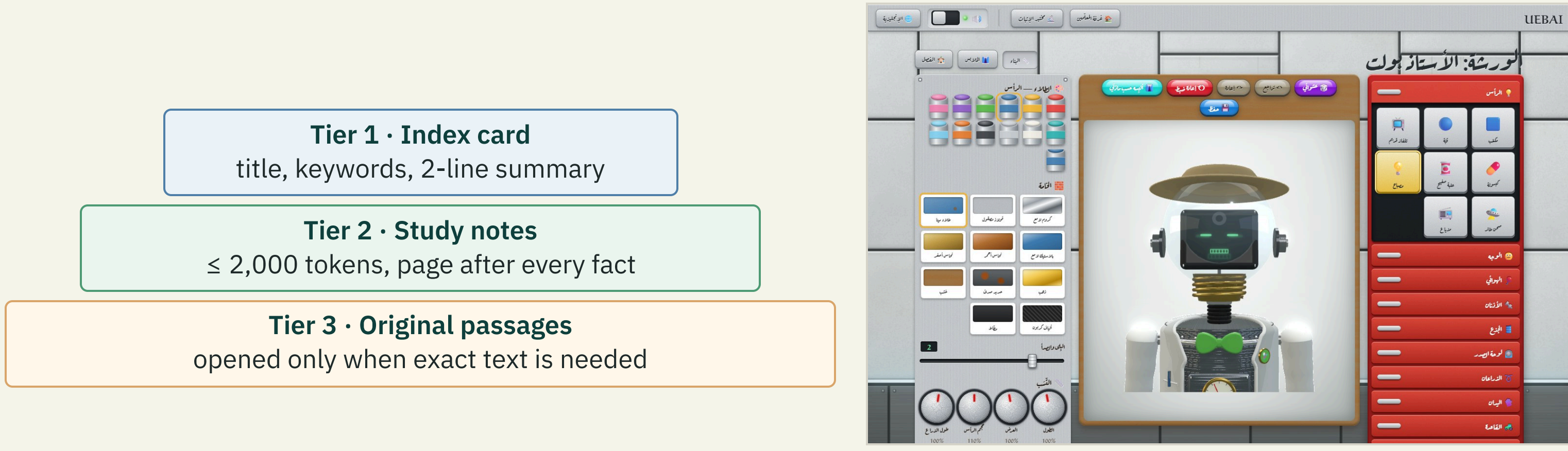


Fig. 3. Students design their own 3D teacher.

Cross-source merging	Teacher slides added: 27 topics merged into book cards; 2 conflicts flagged (fungi: 4 vs 5 phyla)
Verified citations	Every page number is checked against the book
Reviewable cards	Teachers can edit, merge and split cards

Conclusion

Main conclusion

Answering from topic cards cut the text read per question 55 times and made answers start about twice as fast, with answer quality statistically not worse than reading the whole book.

Performance 92% vs 97% key facts over 69 paired runs	Impact ≈55× less text to pay for; ≈3,000 tokens fit even small models	Readiness Working prototype tested on 2 MoE textbooks
--	---	---

Hypothesis (set before testing)	Verdict
H1 Faster: median $\geq 3\times$	Partly 2× overall ($p = 0.024$); $\geq 3\times$ on 2 of 4 models
H2 Cheaper: $\geq 20\times$ fewer tokens	Yes 55× fewer, in 69 of 69 pairs
H3 Quality not worse by > 10 pts	Yes both books together; History alone not yet decided

Limitations

The speed gain varies by model (1.1× to 28×). In History, the keyword router sometimes picks the wrong card (85% vs 87% for plain chunks). No classroom trial or blind teacher rating yet; a biology teacher's 9/10 review is informal. We also predicted that overlapping slides would hurt plain chunks; they did not, and we report that.

Future Work & References

- Blind rating by teachers and a classroom pilot with students.
- A semantic router that opens more cards when topics are close.
- Student and teacher accounts; integration with school platforms.

- Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS.
- Liu, N. F. et al. (2024). Lost in the Middle: How Language Models Use Long Contexts. TACL.
- Sarathi, P. et al. (2024). RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval. ICLR.
- Edge, D. et al. (2024). From Local to Global: A Graph RAG Approach to Query-Focused Summarization. arXiv:2404.16130.
- Saudi Ministry of Education. Biology 1 and History secondary textbooks.

UEBAI PROJECT FILES



github.com/RashedAhmedAseri/UEBAI